

LABUHA OESMAN SADIK, Indonesia

WMO: 974600

Lat: 0.6379S

Lon: 127.5010E

Elev: 30

StdP: 100.97

Time Zone: 9.00 (E09)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	22.0	19.4	14.2	22.4	20.2	14.9	23.3	5.6	28.9	5.0	28.8	0.1	N/A	0.277

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	7.5	32.6	25.7	32.1	25.7	31.7	25.7	26.8	30.4	26.6	30.3	26.4	30.1	2.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.9	21.3	28.8	25.6	20.9	28.7	25.3	20.5	28.5	84.5	30.6	83.4	30.4	82.6	30.3	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.3	4.4	3.7	19.2	34.1	2.4	1.7	17.5	35.3	16.1	36.3	14.7	37.2	13.0	38.4		
(5)			18.9	28.5	1.1	0.9	18.1	29.2	17.5	29.7	16.9	30.2	16.1	30.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.9	27.3	27.1	27.3	27.4	27.2	26.6	26.1	26.0	26.3	26.9	27.3	27.4
	DBStd	0.88	0.73	0.73	0.73	0.67	0.67	0.79	0.75	0.83	0.75	0.66	0.59	0.76
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6168	536	479	535	521	532	498	498	496	489	524	518	540
	CDD18.3	3126	278	246	277	271	274	248	240	238	239	266	268	282
	CDH23.3	28884	2655	2307	2654	2583	2629	2163	1917	1934	2098	2546	2626	2772
	CDH26.7	9404	899	788	933	876	877	609	477	485	608	892	970	990
(14) Wind	WSAvg	1.0	0.9	0.9	0.9	0.9	0.9	0.9	0.9	1.0	1.1	1.2	0.9	0.9
(15) Precipitation	PrecAvg	1964	215	210	198	184	171	132	129	117	107	132	168	201
	PrecMax	3542	372	422	319	303	273	305	334	306	291	345	370	376
	PrecMin	1164	73	72	44	51	2	2	0	0	0	34	0	0
	PrecStd	690	91	102	78	68	83	102	113	112	98	110	95	97
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.6	32.6	32.7	32.8	32.2	31.7	31.1	31.2	31.7	32.4	32.7	33.1
		MCWB	25.5	25.7	25.5	25.8	25.9	26.0	25.3	25.4	25.5	25.3	25.8	25.7
	2%	DB	31.8	31.9	32.1	32.0	31.7	31.1	30.3	30.4	30.9	31.7	32.1	32.4
		MCWB	25.7	25.5	25.5	26.0	26.0	25.9	25.3	25.2	25.3	25.5	25.6	25.8
	5%	DB	31.3	31.3	31.6	31.4	31.2	30.4	29.8	29.8	30.2	31.1	31.6	31.7
		MCWB	25.7	25.5	25.5	26.0	26.0	25.8	25.2	25.0	25.2	25.6	25.7	25.8
	10%	DB	30.8	30.7	30.9	30.8	30.6	29.8	29.1	29.2	29.7	30.5	31.0	31.0
		MCWB	25.6	25.4	25.5	25.9	26.0	25.7	25.1	25.0	25.1	25.5	25.8	25.8
	0.4%	WB	26.7	26.6	26.7	26.9	27.1	26.9	26.2	26.2	26.3	26.8	27.2	27.1
		MCDB	30.2	30.7	30.7	30.7	30.2	30.3	29.7	29.4	29.9	30.5	31.0	30.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	26.3	26.2	26.3	26.6	26.7	26.5	26.0	25.8	26.1	26.4	26.6	26.6
		MCDB	30.1	30.2	30.3	30.4	30.0	29.8	29.2	29.0	29.6	30.1	30.5	30.5
	5%	WB	26.1	25.9	26.0	26.2	26.4	26.2	25.6	25.5	25.7	26.1	26.2	26.2
		MCDB	30.0	29.9	30.1	30.0	29.8	29.4	28.7	28.7	29.1	29.9	30.1	30.2
	10%	WB	25.8	25.6	25.7	26.1	26.2	25.9	25.4	25.2	25.4	25.7	25.9	26.0
		MCDB	29.6	29.4	29.5	29.8	29.6	29.0	28.2	28.2	28.7	29.4	29.8	29.9
(35) Mean Daily Temperature Range	5% DB	MDBR	7.2	7.4	7.6	7.3	7.2	6.5	6.5	6.5	7.1	7.9	8.0	7.5
		MCDBR	8.0	8.3	8.6	8.2	8.1	7.5	7.6	7.3	7.9	8.6	8.8	8.6
		MCWBR	3.3	3.4	3.5	3.5	3.7	3.6	3.6	3.5	3.8	4.0	3.8	3.5
	5% WB	MCDBR	7.4	7.5	7.8	7.5	7.3	6.8	6.7	6.6	7.2	7.9	8.1	7.7
		MCWBR	3.3	3.3	3.5	3.4	3.6	3.5	3.5	3.4	3.6	3.9	3.9	3.6
(40) Clear-Sky Solar Irradiance	taub		0.425	0.433	0.427	0.415	0.404	0.406	0.409	0.416	0.426	0.458	0.434	0.430
	taud		2.465	2.431	2.450	2.486	2.537	2.543	2.537	2.500	2.459	2.341	2.416	2.439
	Ebn at Noon		907	902	899	889	875	860	864	876	886	869	891	897
	Edh at Noon		116	122	119	112	102	100	101	108	116	132	121	118
(44) All-Sky Solar Radiation	RadAvg		5.26	5.50	5.61	5.44	5.30	4.69	4.79	5.16	5.49	5.81	5.54	5.26
	RadStd		0.29	0.41	0.44	0.28	0.36	0.41	0.64	0.73	0.62	0.42	0.35	0.40

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page